Institute for
Data Science Foundations
E-21

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Lecture 09

In the last lecture, we studied Markov properties that are associated with a graphical model. Given a graph $G = (V, E)$, the elements of the corresponding graphical model have the form

$$p = \frac{1}{Z} \exp\left(\sum_{\substack{C \text{ clique} \\ w.r.t.\ G}} f_C \right), \tag{1}$$

where f_C only depends on $x_i, i \in C$, that is

$$f(x_C, x_{V \setminus C}) = f(x_C, x'_{V \setminus C})$$

for all x_C, $x_{V \setminus C}$, $x'_{V \setminus C}$.

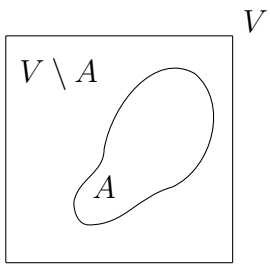
In this lecture, we study an algorithm that allows us to sample from (1). This algorithm is called the Gibbs-sampling algorithm because the probability measure (1) is knows from physics as Gibbs distribution and the sampling mimics some physical process.

Sampling in large spaces is a challenge. For a coin or dice we have two and six possible outcomes, respectively. If we want to sample a 100×100 pixel black/white-image then we are selecting an image from $2^{10.000}$ possible images that is typical with respect to some probability measure p. This corresponds to a dice with $2^{10.000}$ instead of six possible outcomes. How can we realise such a dice? The solution is by not doing it at once. One breaks down the process of sampling from this huge space into many processes that are simple to implement. This way, spatial complexity is translated to temporal complexity.

Now we come to two central ideas underlying the method of Gibbs-sampling.

Idea 1: Compute conditional probabilities for the Gibbs measure.

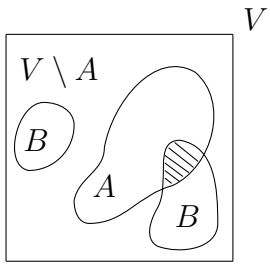
$$p(x) = \frac{1}{Z} \exp\left(\sum_{B \subseteq V} f_B(x) \right)$$



$$\begin{aligned} p(x_A, x_{V \setminus A}) &= \frac{p(x_A, x_{V \setminus A})}{p(x_{V \setminus A})} \\ &= \frac{p(x_A, x_{V \setminus A})}{\sum_{x'_A \in X_A} p(x'_A, x_{V \setminus A})} \\ &= \frac{\frac{1}{Z} \exp\left(\sum_{B \subseteq V} f_B(\overset{x_A x_{V \setminus A}}{x}) \right)}{\sum_{x'_A} \frac{1}{Z} \exp\left(\sum_{B \subseteq V} f_B(x'_A, x_{V \setminus A}) \right)} \end{aligned}$$

$$= \frac{\exp\left(\sum_{\substack{B\subseteq V\\ B\cap A\neq\emptyset}} f_B(x_A, x_{V\setminus A}) + \sum_{\substack{B\subseteq V\\ B\cap A=\emptyset}} f_B(x_A, x_{V\setminus A})\right)}{\sum_{x'_A} \exp\left(\sum_{\substack{B\subseteq V\\ B\cap A\neq\emptyset}} f_B(x'_A, x_{V\setminus A}) + \sum_{\substack{B\subseteq V\\ B\cap A=\emptyset}} f_B(x'_A, x_{V\setminus A})\right)}$$

$$= \frac{\exp\left(\sum_{\substack{B\subseteq V\\ B\cap A\neq\emptyset}} f_B(x_A, x_{V\setminus A})\right)}{\sum_{x'_A} \exp\left(\sum_{\substack{B\subseteq V\\ B\cap A\neq\emptyset}} f_B(x'_A, x_{V\setminus A})\right)}$$



Observation: Many terms cancel out so that sampling x_A given $x_{V\setminus A}$ becomes much simpler than sampling the full configuration $x_A, x_{V\setminus A} = x$. We will concentrate on the case where A consists of only one node.

Core Example.

$$p(x) = \frac{1}{Z} \exp\left(\sum_{s\sim t} \theta_{st} x_s x_t + \sum_s \theta_s x_s\right)$$

We choose one element r, that is $A = \{r\}$, and repeat the previous calculation.

$$
\begin{aligned}
p(x_r|x_{V\setminus r}) &= \frac{\exp\left(\sum\limits_{\substack{s\sim t\\ r\notin\{s,t\}}}\theta_{st}x_sx_t + \sum\limits_{\substack{s\\ r\neq s}}\theta_s x_s + \sum\limits_{s\sim r}\theta_{sr}x_sx_r + \theta_r x_r\right)}{\sum\limits_{x'_r}\exp\left(\sum\limits_{\substack{s\sim t\\ r\notin\{s,t\}}}\theta_{st}x_sx_t + \sum\limits_{\substack{s\\ r\neq s}}\theta_s x_s + \sum\limits_{s\sim r}\theta_{sr}x_sx'_r + \theta_r x'_r\right)}\\
&= \frac{\exp\left(\sum\limits_{s\sim r}\theta_{sr}x_sx_r + \theta_r x_r\right)}{\sum\limits_{x'_r}\exp\left(\sum\limits_{s\sim r}\theta_{sr}x_sx'_r + \theta_r x'_r\right)}
\end{aligned}
$$

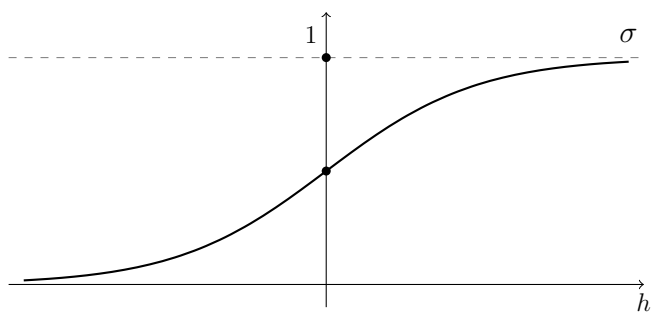
$$
\begin{aligned}
&\overset{x_s\in\{-1,+1\}}{\overset{\downarrow}{=}} \frac{\exp\left(\sum\limits_{s\sim r}\theta_{sr}x_sx_r + \theta_r x_r\right)}{\exp\left(\sum\limits_{s\sim r}\theta_{sr}x_sx_r + \theta_r x_r\right) + \exp\left(-\sum\limits_{s\sim r}\theta_{sr}x_sx_r - \theta_r x_r\right)}\\
&= \frac{1}{1+e^{-2x_r\left(\sum\limits_{s\sim r}\theta_{sr}x_s+\theta_r\right)}}
\end{aligned}
$$

$\Rightarrow$

$$
\begin{aligned}
p\left(x_r = +1|x_{V\setminus\{r\}}\right) &= \frac{1}{1+e^{-2\left(\sum\limits_{s\sim r}\theta_{sr}x_s+\theta_r\right)}}\\
&= \sigma\left(\sum_{s\sim r}\theta_{sr}x_s + \theta_r\right) \qquad (2)
\end{aligned}
$$

with

$$\sigma(h) = \frac{1}{1+e^{-2h}}$$



Here, we have only two possible outcomes, $x_r = -1$, $x_r = +1$, basically a coin, and the probability for $x_r = +1$ is given by (2). ★

Now we come to the second central idea underlying the method of Gibbs-sampling.
Idea 2: Use the simple rules for the conditional probabilities as a builing block of a temporal

process. Here, we use the conditional probability to define a Markov kernel/stochastic matrix according to which a new x' is generated at time $n+1$ if we have x at time n:

$$P_A(x;x') := \begin{cases} p(x'_A|x_{V\setminus A}), & \text{if } x_t = x'_t \text{ for all } t \in V \setminus A \\ 0, & \text{otherwise.} \end{cases}$$

This simply means that the new x' is obtained from the current x by only changing the values x_i for $i \in A$, which is done according to the conditional probabilities $p(x'_A|x_{V\setminus A}), x'_A$.

Core Example continued:
For $r \in V$, we leave the values for all $s \neq r$ unchanged and update only x_r by a new x'_r such that $x'_r = +1$ with probability

$$\frac{1}{1+e^{-2(\sum_{s\sim r}\theta_{sr}x_s+\theta_r)}}$$

★

Now we combine the two ideas, Idea 1 and Idea 2, in order to define the Gibbs-sampling. For that we enumerate V by $\{1,2,\cdots,N\}$.

- Initialisation: We begin with an arbitrary x. We can obtain such an x, for instance, by sampling all x_i independently.
- We update the value x_r of the currently chosen r in terms of the local update rule.
- After having updated the values of all nodes $1,2,\cdots,N$, we iterate this by starting again with node 1, going to node 2, etc.
- After cycling through all nodes several times, the x will be more and more typical for the Gibbs-measure.

We finally state the theorem that underlies Gibbs-sampling.

Theorem. *(Geman and Geman 1984) Let P be the Markov kernel/stochastic matrix obtained from a Gibbs-distribution p. For all initial probability measures q one has*

$$\lim_{n\to\infty} qP^n = p$$

uniformly in q. In particular, p is the only stationary distribution of P.